AI Ethics – AI labor

What is the microwork behind AI solution?

1. Introduction: digital platform labor in platform economy

1.1. Platform and triangulation process

What is a platform?

Intermediary firm based on socio-technical devices that produce **matches** between people, info, goods/services and **captures data** from these interactions ⟶ it **taskifies** contributions to make them interoperable

- Algorithmic match-making across user groups (workers/client, riders/drivers, hots/guests)
- Value capture from content, data, metadata generated by users
- Put “users” at work organizing unpaid/underpaid tasks ⟶ digital labor continuum
- “Taskification” to generate data and enable coordination
- No classic Taylorism: coordination outside the firm across multitudes of users

Platform triangulation process

Abel & Dieuaide (2023). Dual workspace and new form of subordination. The case of location-based platforms, *Socio-Anthropo*

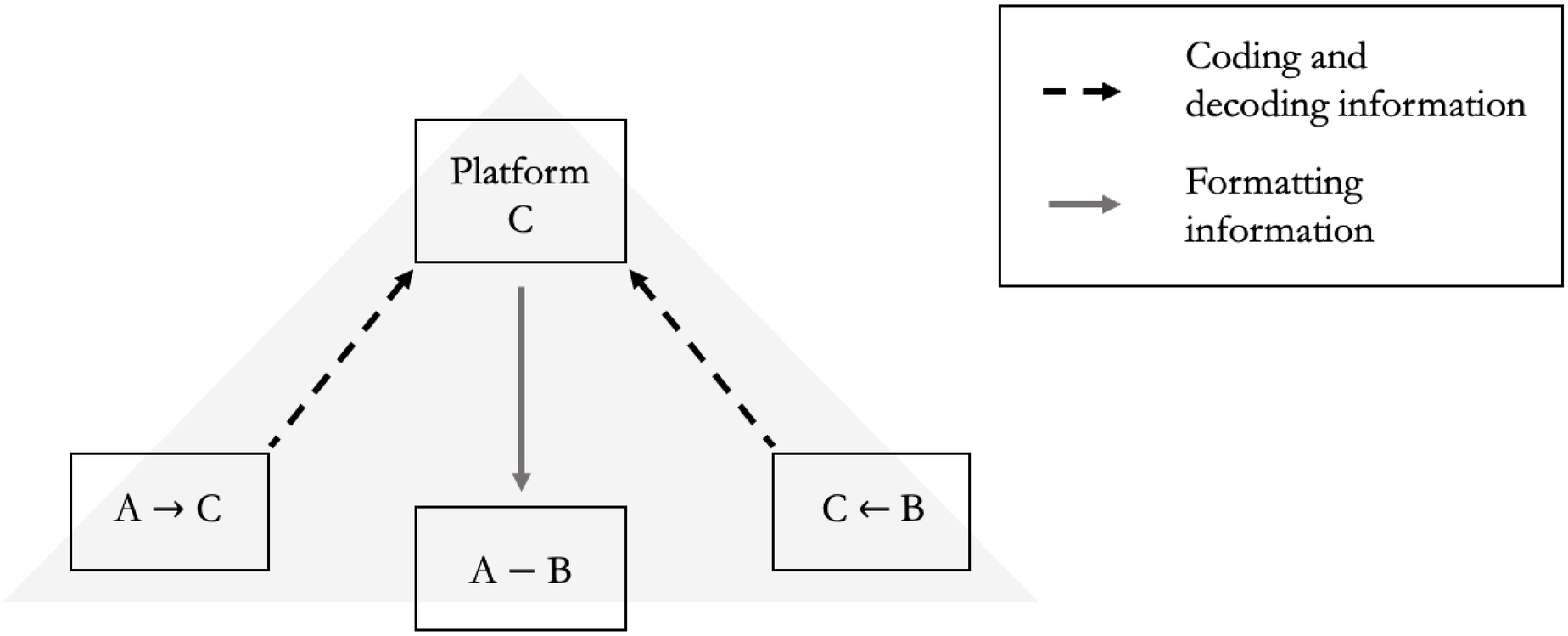


Fig. 1. The platform triangulation process

1.2. What is digital labor?

What is digital labor?

Digital labour spans **unpaid** user activities to **micro-paid** piecework and **freelancing** outside regular

- Terranova's « **free labor on the net »**: building sites, coding, mailing lists, virtual spaces (origin).
- Beyond pro-bono: filling profiles, curating friend lists, posting/rating ⟶ **monetizable by-products**
- Continuum ⟶ invisible/informal arrangements lacking most of workers rights (paid leave, retirement, event to right to be paid)



Why this matters for AI Ethics?

- AI solution depends on extracting data/metadata ⟶ fuel for automation
- AI solution development displaces rather than replaces work ⟶ proliferation of micro-workers and « non-workers »
- Who are they? Where? Under which rules and apay?

2. AI and intermediation

2.1. Who are the clients?

Who are the clients?

AI producers : Big Tech, start-ups, corporate R&D, and labs ⟶ all depend on **data-hungry ML** (esp. supervised & deep learning).

Recurring needs

- Label quality > volume
- Domain coverage and local specificity
- Lifecycle continuity (up-to-date data)
- Accuracy–cost–time trade-offs under safety constraints
- Governance, secrecy, and provenance
- Risk containment (legal, reputational, operational)

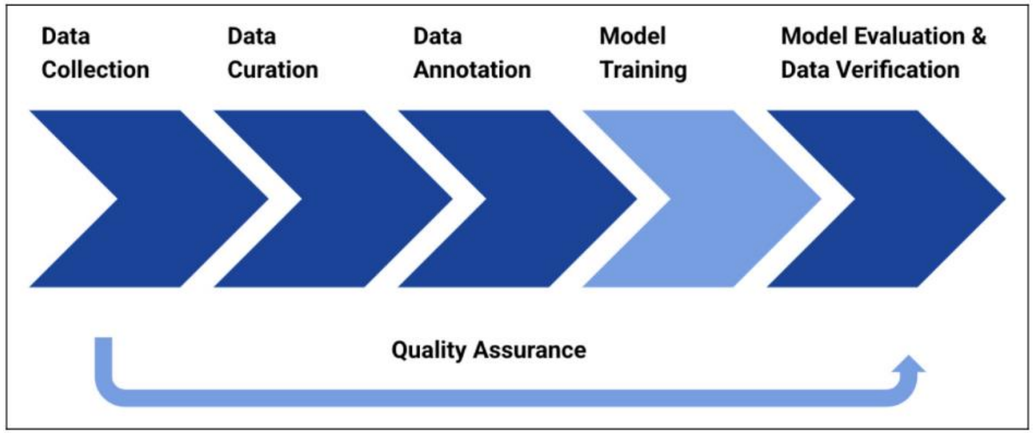


Fig. 2. The AI data pipeline

2.2. Plurality of platforms

Who are the AI platforms?

Who are the AI platforms?

amazon beta mechanical turk

microWorkers
work & earn or offer a micro job

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Typology of AI-data institutions

Muldoon et al. (2024). A typology of artificial intelligence data work, *Big Data & Society*

Two axes:

- Employment relationships: crowdworker or employees
- Service focus: external vs internal; generalist vs AI specific

Table 1. AI data work institutions.

			Type of worker	
			Crowdsourced/self-employed	*Employees*
Type of service	***External***	*Generalist company*	*Type A*: Generalist platform	*Type B*: Generalist BPO
		AI data company	*Type C*: AI data platform	*Type D*: AI data BPO
	Internal	*AI company*	*Type E*: Internal data platform	*Type F*: Internal data services

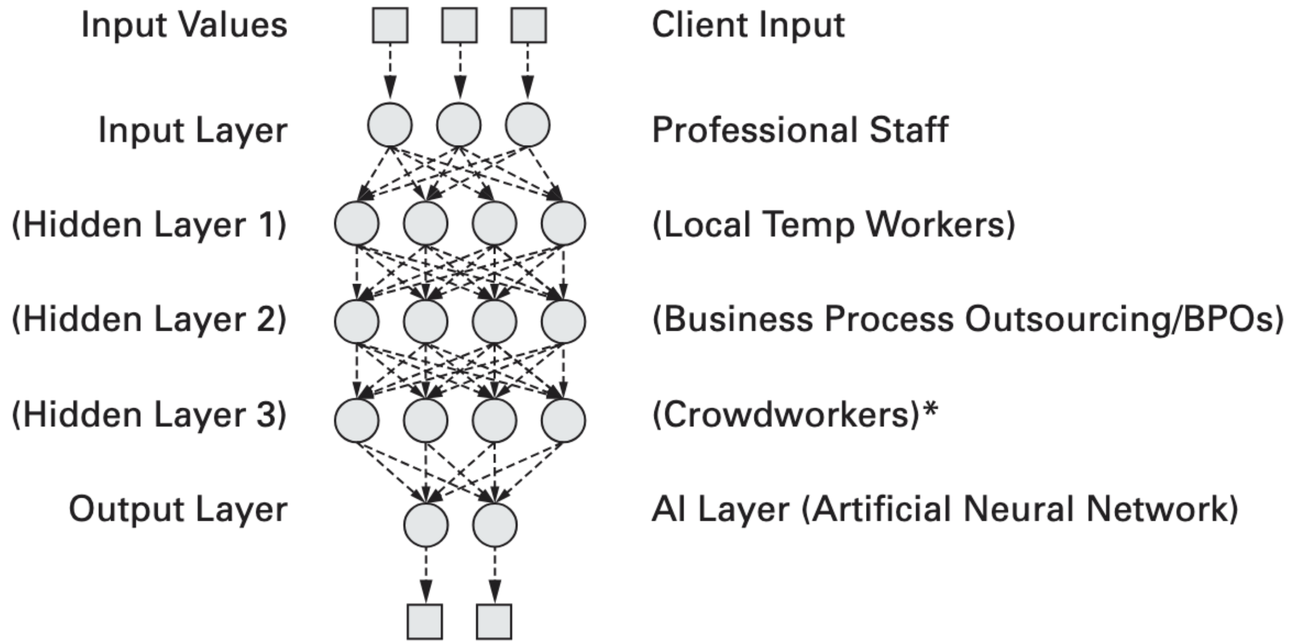
AI: artificial intelligence; BPO: business process outsourcing.

Source: Muldoon et al. 2024

Schmidt (2022). The Planetary Stacking Order of Multilayered Crowd-AI Systems Open Access in *Digital Work in the Planetary Market,* ed. Graham & Ferrari, MIT Press

To hit **95–99%+** accuracy and tight service agreement, intermediaries evolved from flat marketplaces to **multi-layer stacks** (tooling + curated labour + internal QA).

- Structural self-similarity": organisational hidden layers mirror technical hidden layers (input → hidden → output), yielding **"black boxes within black boxes."**
- Result: better accuracy & turnaround, but rising **opacity** on provenance and labour conditions



* Because the crowdworkers, in a form of suboutsourcing, rent out their accounts to others, they, too, become a hidden layer or black box, not just for the clients, but effectively also for the platform providers.

Source: Schmidt (2022)

3. AI micro work

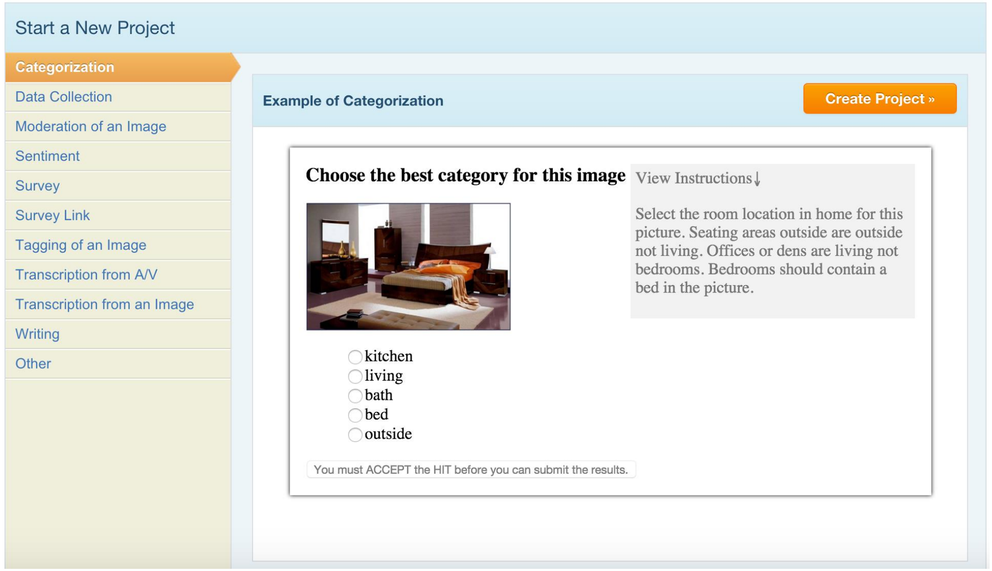


Fig. AMT worker screen capture

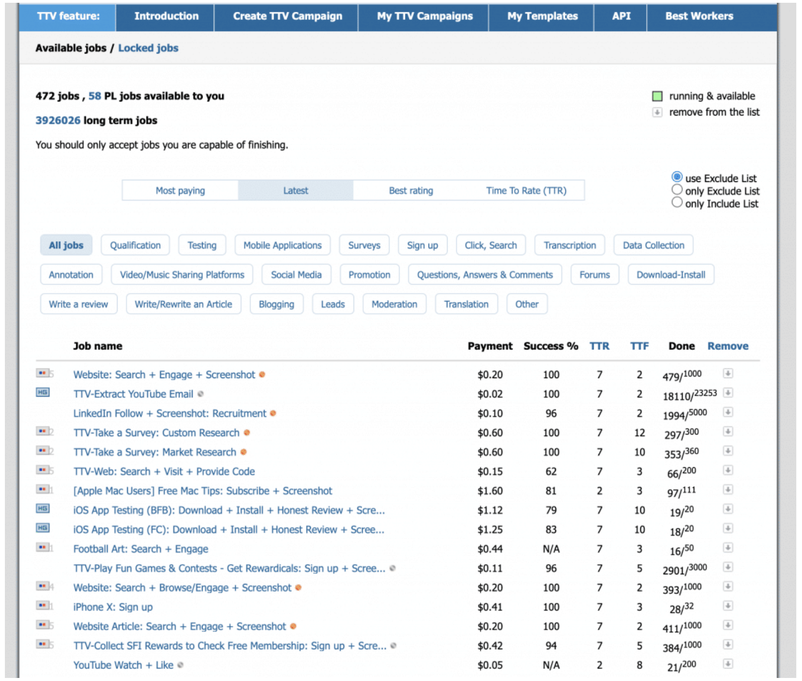


Fig.Microworkers worker screen capture

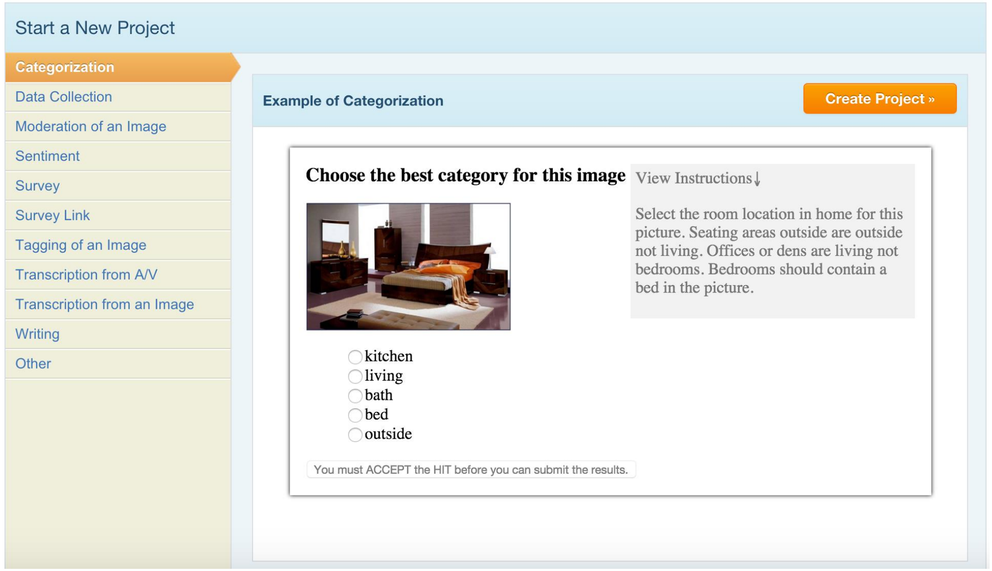


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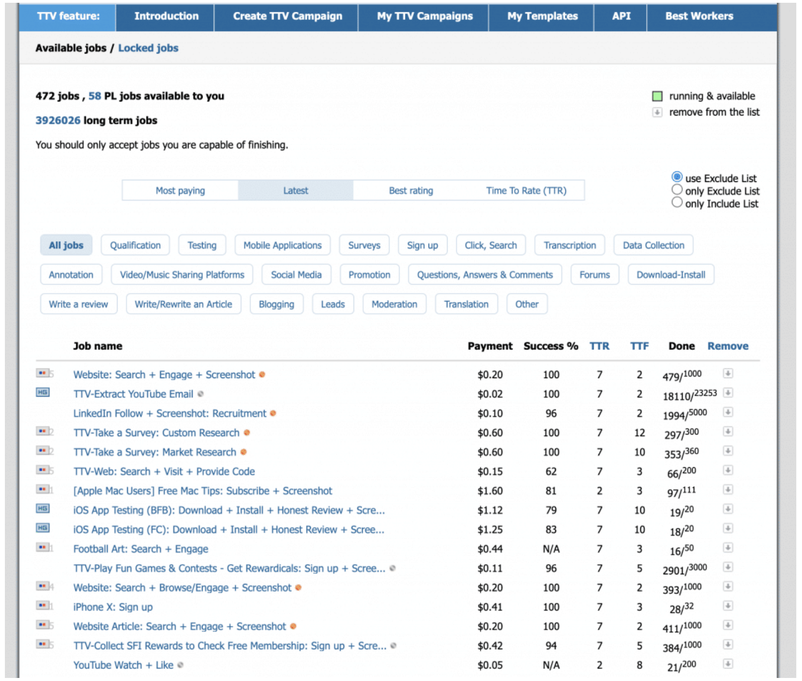


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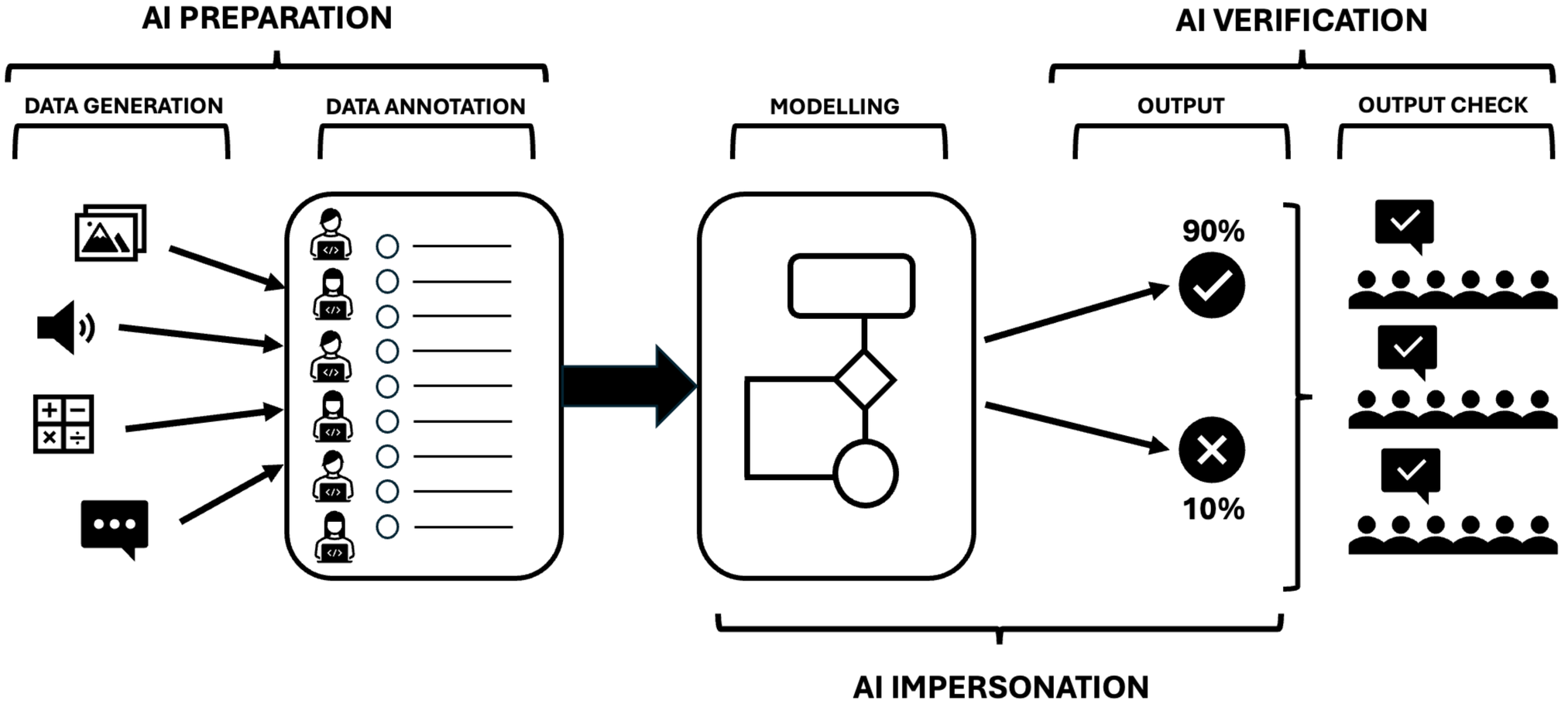
How are workers “in the loop”?

- AI data work = human labour required to support ML through **preparation** and **evaluation** of datasets and model outputs, often outsourced to low-paidand marginalised workers (Muldoon et al. 2024)

- **Three persistent roles: Preparation** (pre-robot), **Impersonation** (humans stand in), **Verification** (post-robot checks feeding retraining) (Tubaro et al. 2020)

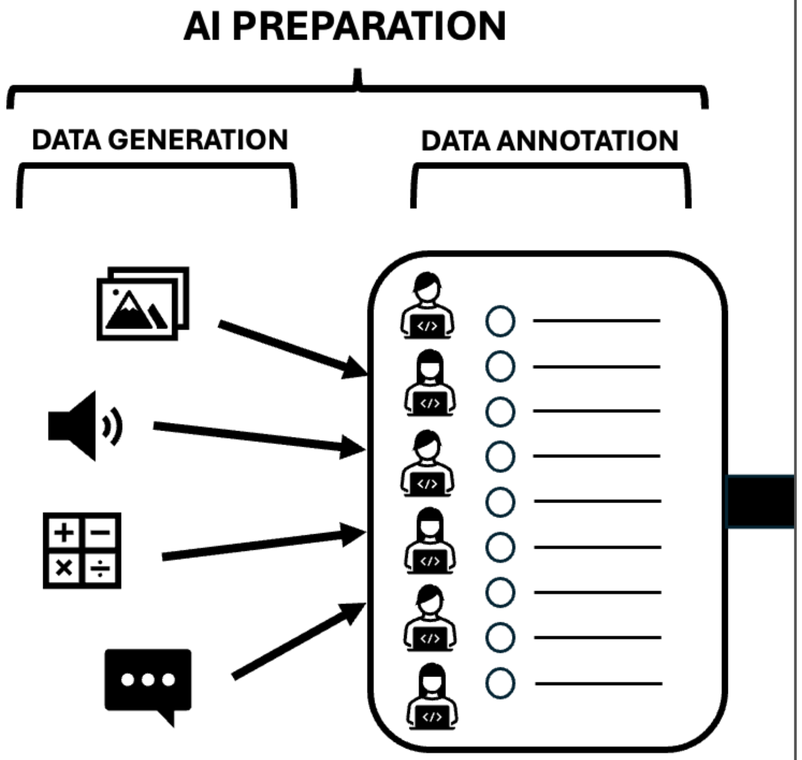
3.1.AI preparation

Tubaro et al. (2020). The trainer, the verifier, the imitator: Three ways in which human platform workers support artificial intelligence, *Big Data & Society*



Source: Tubaro et al. 2020

Tubaro et al. (2020). The trainer, the verifier, the imitator: Three ways in which human platform workers support artificial intelligence, *Big Data & Society*

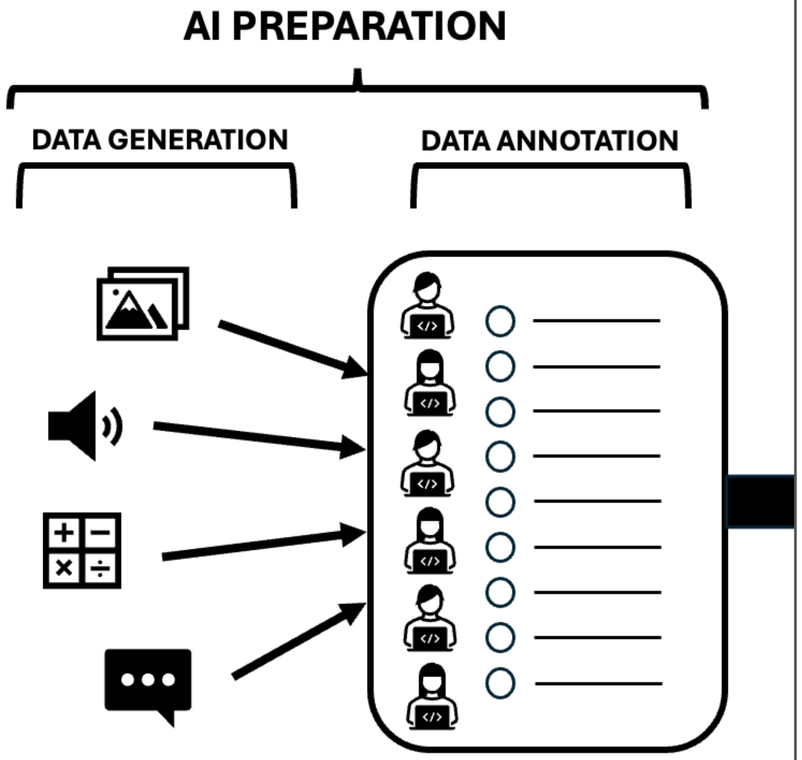


Preparation = data generation + annotation

- **Data generation:** collect voices/images/text (accents, slang, noisy contexts; country-specific corpora).
- **Annotation:** from **classification/boxes** to **polygons/landmarks** up to **pixel-accurate semantic segmentation** for safety-critical domains.

Source: Tubaro et al. 2020

Tubaro et al. (2020). The trainer, the verifier, the imitator: Three ways in which human platform workers support artificial intelligence, *Big Data & Society*



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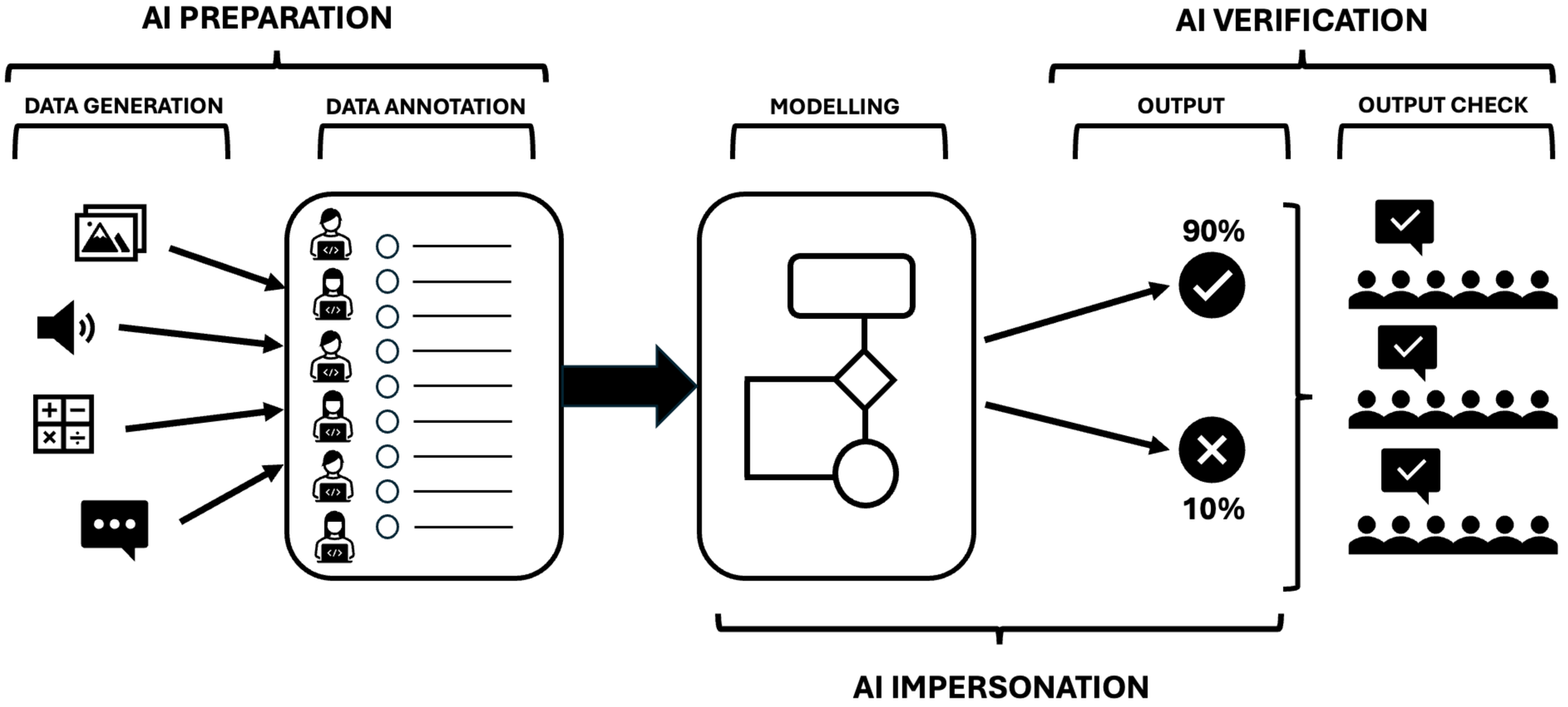
Preparation in practice

- **Voice:** short utterance recording across timbres/accents/noise.
- **Commerce/docs:** match product photos ↔ names across marketplaces; **CV anonymisation** + tagging of name/date/address slots so parsers learn layout (needs **local conventions**).

Source: Tubaro et al. 2020

3.2. AI Impersonation

Tubaro et al. (2020). The trainer, the verifier, the imitator: Three ways in which human platform workers support artificial intelligence, *Big Data & Society*

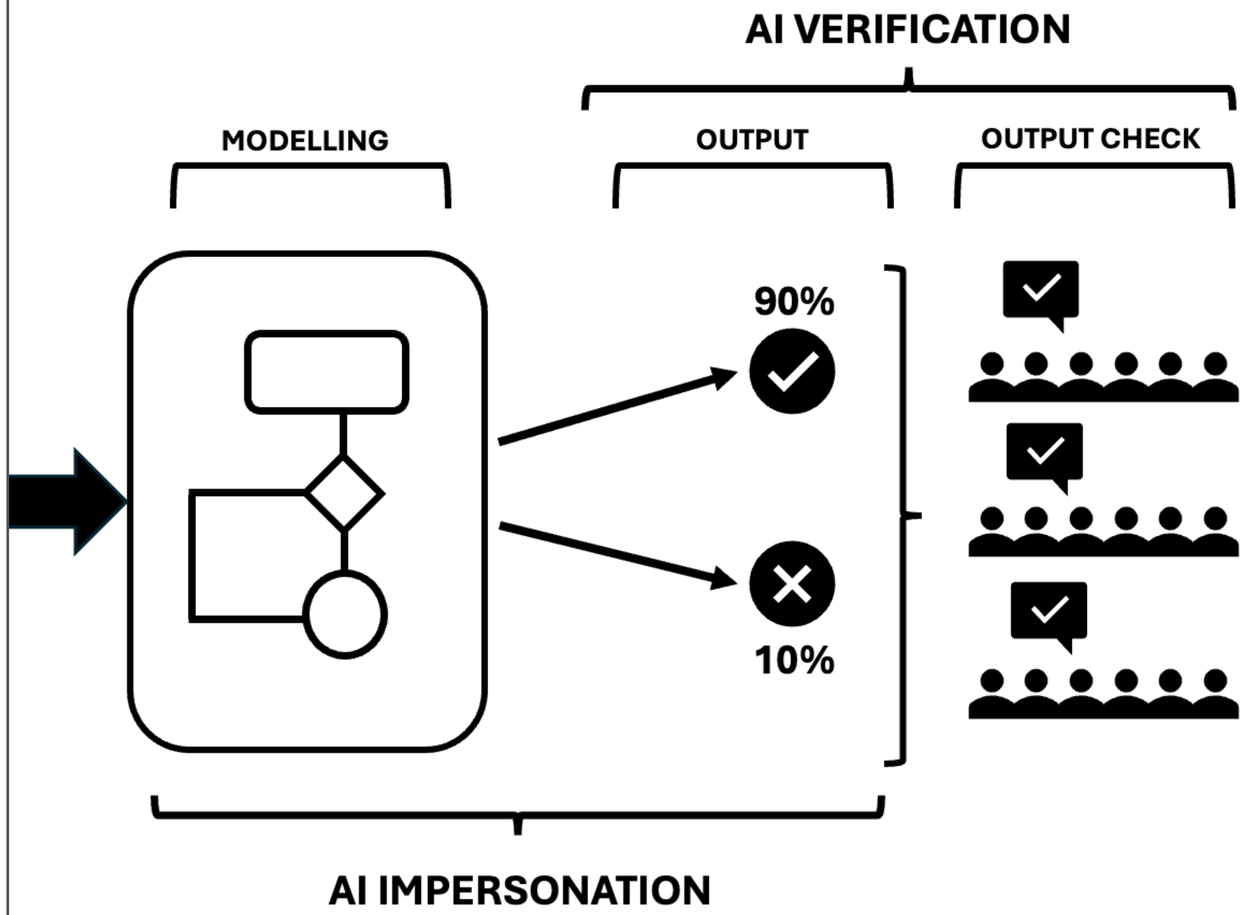


Source: Tubaro et al. 2020

Tubaro et al. (2020). The trainer, the verifier, the imitator: Three ways in which human platform workers support artificial intelligence, *Big Data & Society*

Impersonation: “artificial artificial intelligence”

- **What it is:** humans **partly/fully** substitute for algorithms—by design—to keep services usable and generate training signals.
- **Canonical intuition:** MTurk as “artificial artificial intelligence.”

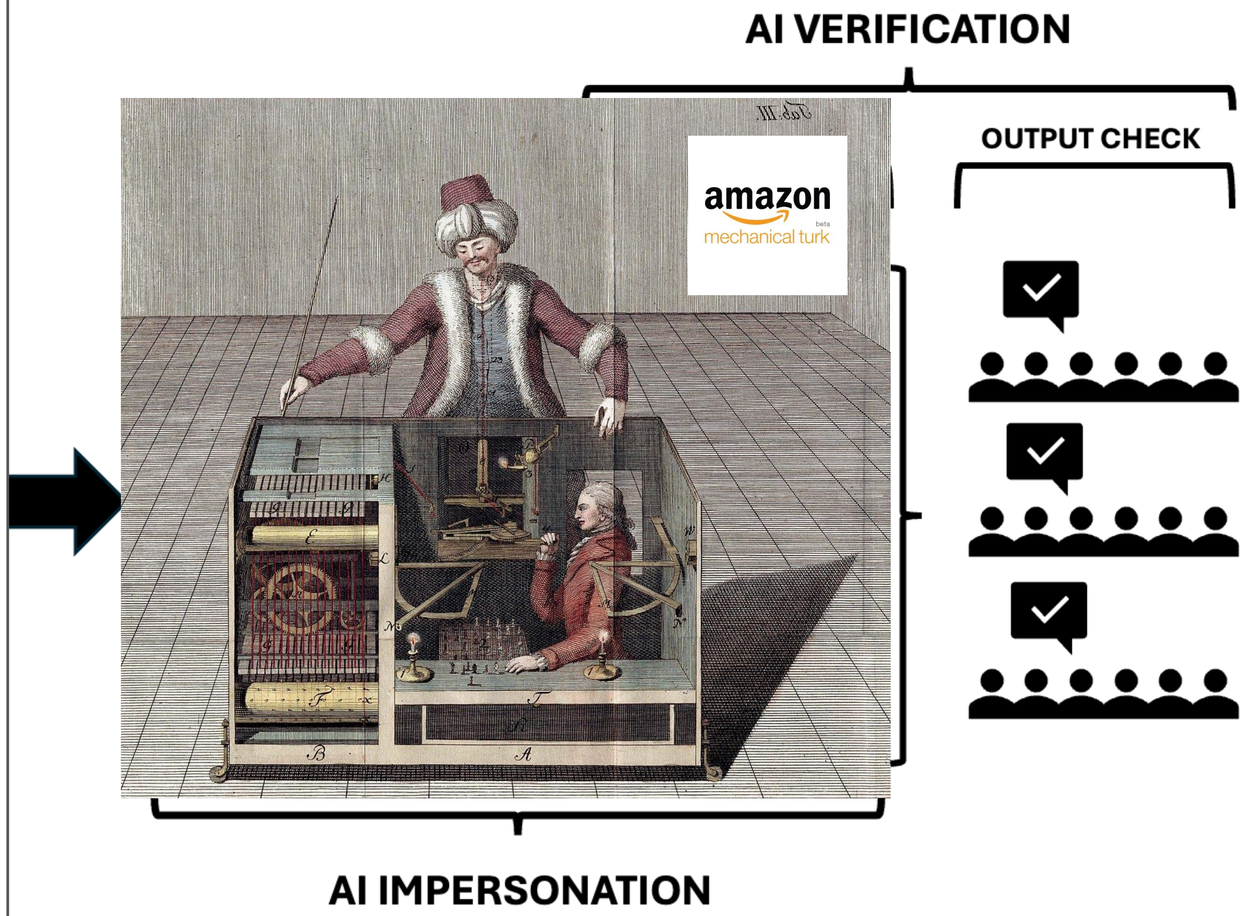


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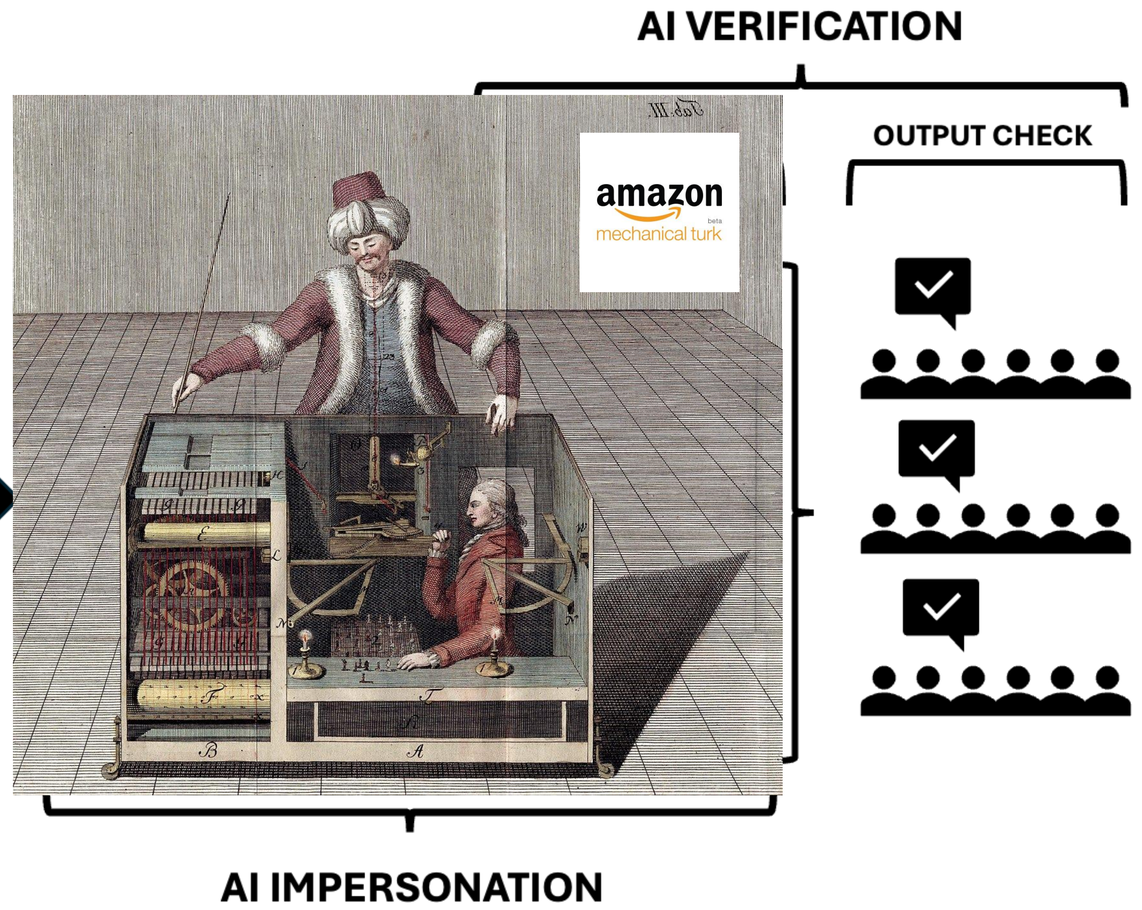


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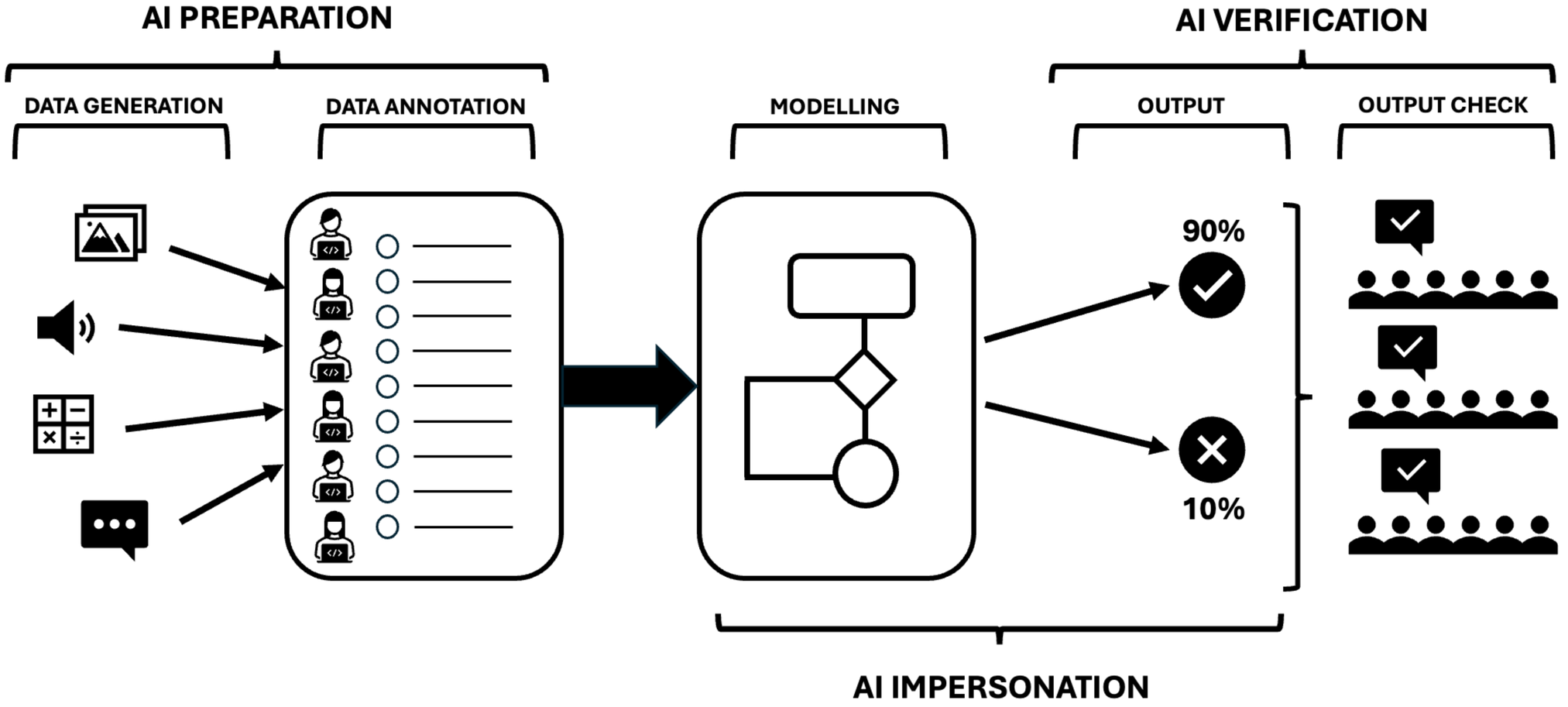
- **What it is:** humans **partly/fully** substitute for algorithms—by design—to keep services usable and generate training signals.
- **Canonical intuition:** MTurk as “artificial artificial intelligence.”
- **Examples: Google Duplex** (as of May 2019, **up to 25% of calls by humans**), **Julie Desk** (“80% machine / 20% human,” with **final human validation**).
- **Why it persists:** gaps in capability; **cost/quality** advantages; stitches prep ↔ verification.



Source: Tubaro et al. 2020

3.3. AI verification

Tubaro et al. (2020). The trainer, the verifier, the imitator: Three ways in which human platform workers support artificial intelligence, *Big Data & Society*

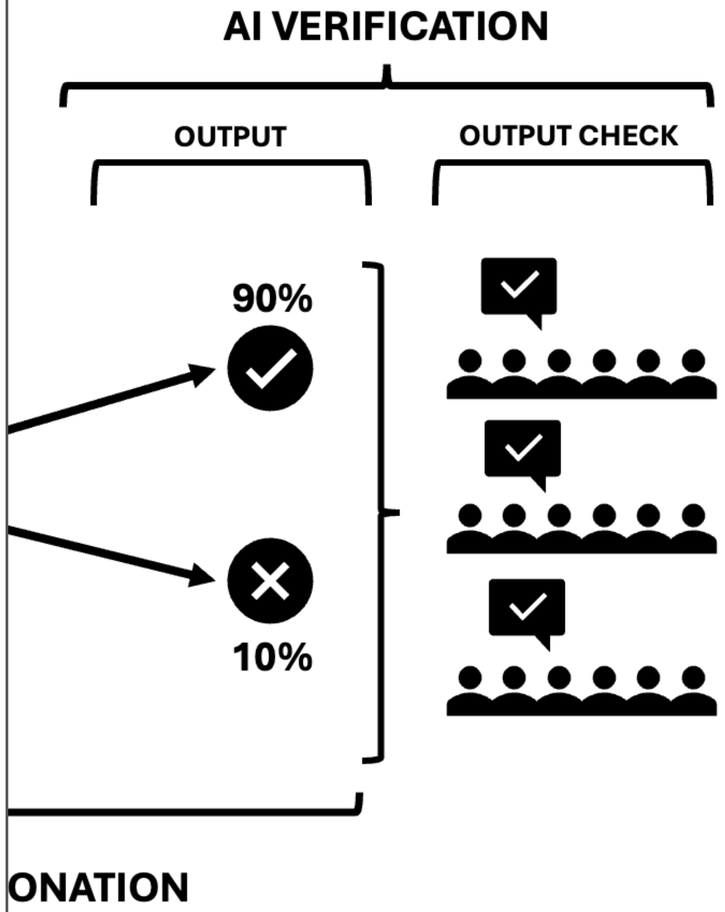


Source: Tubaro et al. 2020

Tubaro et al. (2020). The trainer, the verifier, the imitator: Three ways in which human platform workers support artificial intelligence, *Big Data & Society*

Verification: post-robot quality checks

- **Why it's distinct:** system is **already live**; humans **audit/correct** and produce fresh labels.
- **What workers do:** listen to 3–15s **ASR clips** and correct transcripts (tag noises/confounders); **relevance scoring** for search/assistants; **post-OCR** invoice cleanup.
- **verification never goes away** (any paradigm) and becomes a **recurrent**, strategic need.



Source: Tubaro et al. 2020

4. Sociological approach of AI labor

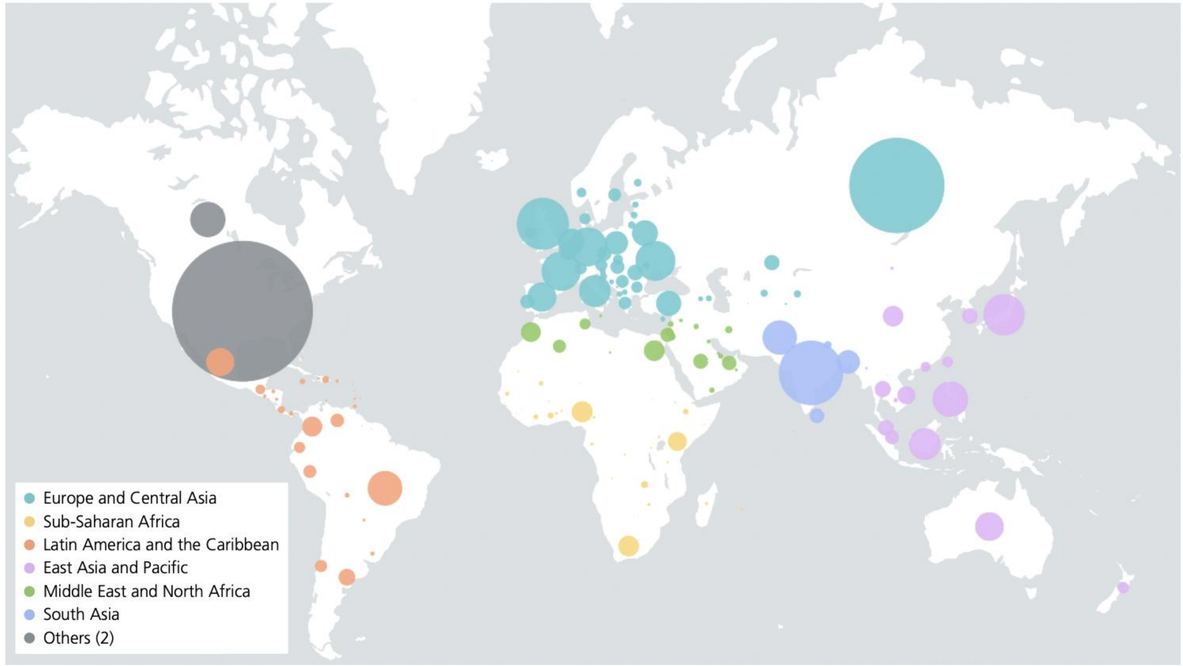
4.1.The Global division of AI labor

A planetary, platform-mediated division of labour

Platforms/BPOs are a **new offshoring institution** matching clients with one-person micro-providers across borders, reshaping where AI data work is done.

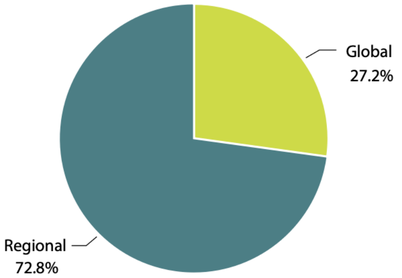
- Matching + low entry barriers for workers in emerging economies.
- Supply concentrates where **language skills, connectivity, wage expectations** align—even if local labour markets are fragile.
- Africa example: English/French-speaking workers enter remote digital work, with **distinct national profiles** and uneven bargaining power.

FIGURE 0.8: Global distribution of online gig work platforms, by traffic



Source: Team database from CB Insights, Pitchbook, and Semrush.
Note: The figure shows the traffic towards gig work platforms, with size depicting magnitude and colors showing different WB Regions.

FIGURE 0.9: Global and regional/local online gig platforms



Source: Team database.

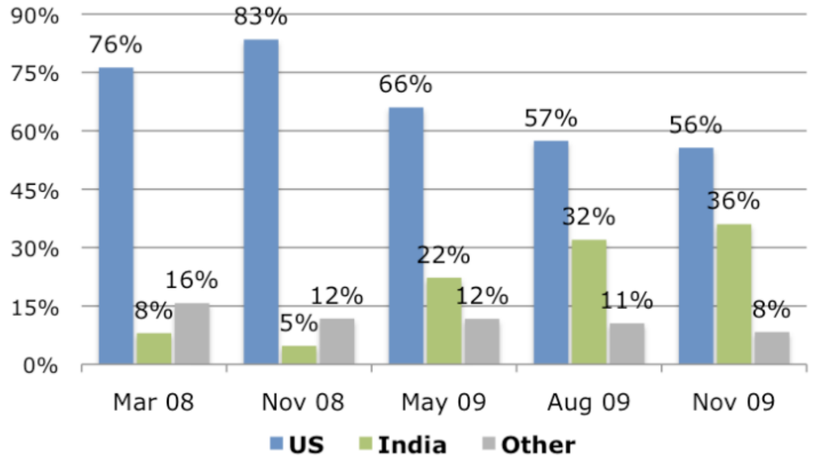
Source: World bank (2023)

From MTurk's early geographies to dispersed hubs

Early crowdwork showed **US + India dominance**; demographics shifted as the market internationalised.

- Early sign: more **young, well-educated Indian** workers using Turking as a **full-time survival strategy** (Ross et al. 2018).
- **Top-20 origin countries** on MTurk: US ≈ **75%**, India ≈ **16%** (2015–2017 panel).
- Registered MTurk workers **>100k (100–200k)**; **~2k active at any time**; **half-life 12–18 months** → constant refresh.

Figure 1. Nationality of MTurk workers over time. Countries with more than 1% of respondents include Canada, the U.K., and the Philippines. We do not have country data for Feb. 2009.



Source: Ross et al. (2010)

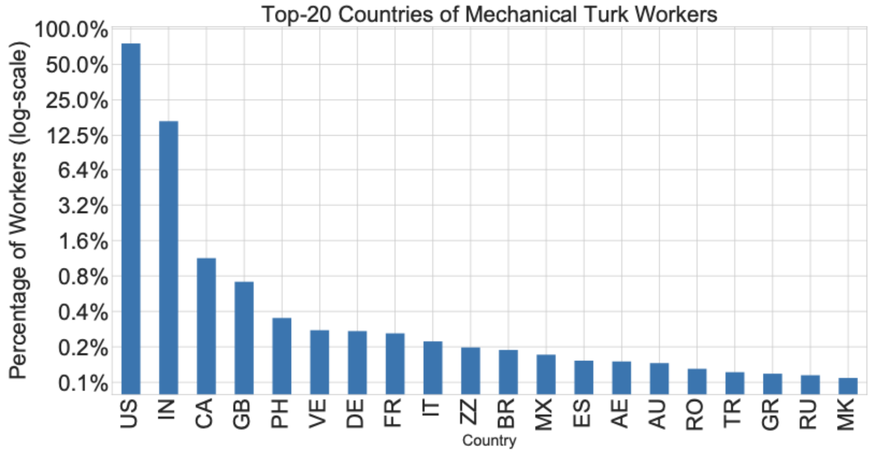


Figure 1: The top-20 countries of origin for Mechanical Turk workers. Most of the workers are from the USA (75%), with India (16%) being second, followed by Canada (1.1%), Great Britain (0.7%), Philippines (0.35%), and Germany (0.27%).

Source: Difallah et al. (2018)



Figure 4: The average percentage of workers from various countries, over time. For US and India, we also show the the 25% and 75% percentiles to illustrate the typical noise levels in our survey measurements.

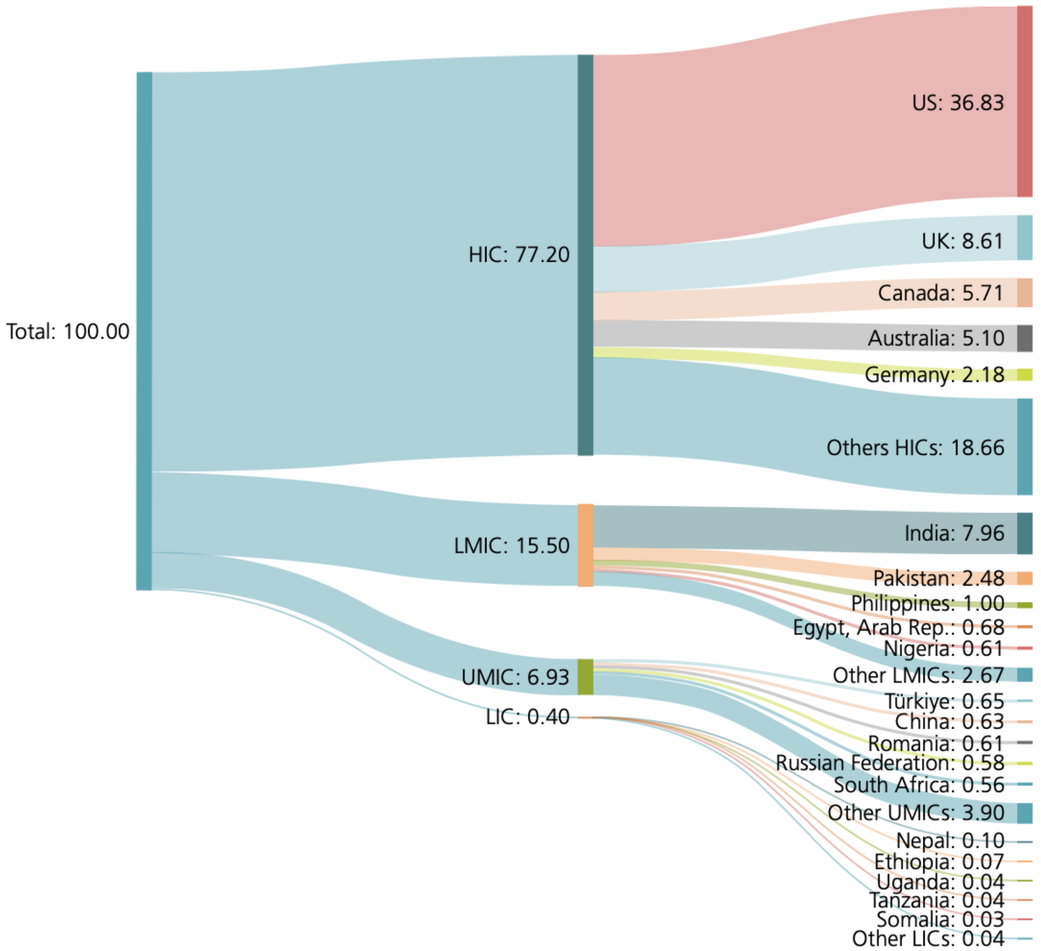
Source: Difallah et al. (2018)

How big is online work (and where does AI fit)?

Global online work estimates: **154–435 million people** (all categories); AI data work is a **sub-segment** within this spatial logic.

- World Bank (multi-country traffic/scraping + survey): **154–435M** online workers (**~4.4–12.5%** of global workforce).
- AI data work follows the same **relocation to lower-income regions**
- **Demand** clusters in North America/Europe;
- **supply** clusters in South Asia, Eastern Europe, Latin America, parts of Africa.

FIGURE 0.19: Demand for online labor, by country and country income groups—2022



Source: World Bank illustration based on Online Labour Index data.
Note: HIC = high-income countries; LIC = low-income countries; LMIC = lower-middle-income countries; UK = United Kingdom; UMIC = upper-middle-income countries; US = United States.

Source: World bank (2023)

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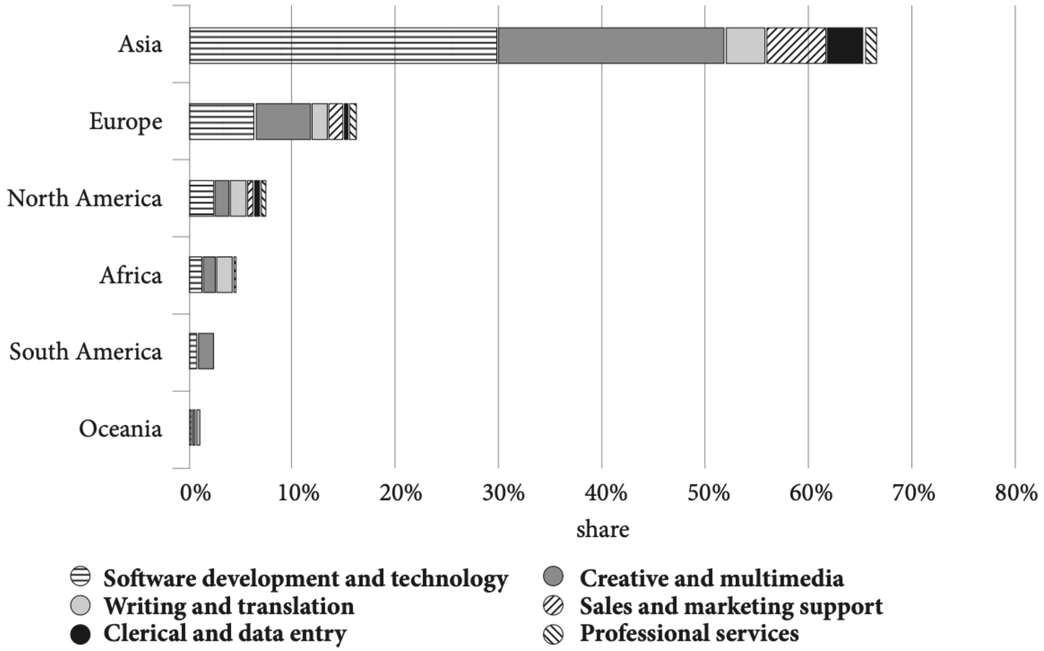
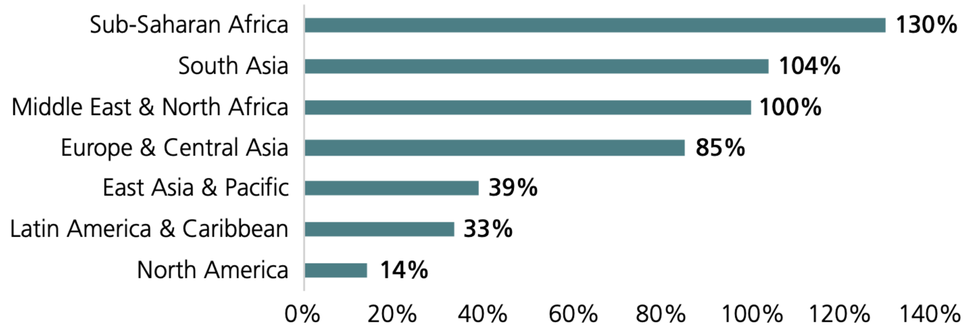


Fig. 3.9 Regional distribution of labour supply on work platforms

Source: Anwar & Graham (2022)

FIGURE 0.20: Growth rate of job postings on one of the largest digital labor platforms for 2016–20, by region



Source: World Bank illustration based on data shared by the Online Labour Index team.

Source: World bank (2023) Anwar & Graham

Table 3.3 Labour supply in selected countries on Upwork on 21 June 2019.

Country	*Potential Workforce*	*Min. $1 earned*	*No earnings*	*Over supply (%)*
Global	**2,302,902**	**196,538**	**2,106,364**	91.4
United States	758,845	40,214	718,631	94.7
India	271,460	32,692	238,768	87.9
Philippines	179,689	24,339	155,350	86.4
Eastern Europe	N/A	N/A	N/A	N/A
Ukraine	69,076	12,531	56,545	81.8
Serbia	31,109	4,169	26,940	86.5
Romania	20,842	2,087	18,755	89.9
South and Central America	N/A	N/A	N/A	N/A
Brazil	27,169	1,490	25,679	94.5
Argentina	14,337	1,475	12,862	89.7
Mexico	14,709	1,140	13,569	92.2
Colombia	10,317	824	9,493	92
Africa	**120,345**	**7,024**	**113,320**	**94.1**
Egypt	44,270	2,091	42,179	95.2
Kenya	21,412	1,616	19,796	92.4
South Africa	15,474	957	14,517	94.1
Nigeria	9,499	827	8,672	91.2
Morocco	9,041	376	8,665	95.8
Tunisia	6,424	270	6,151	95.7
Algeria	4,383	165	4,218	96.2
Ghana	1,799	94	1,705	94.7
Uganda	1,311	54	1,257	95.8

Source: Authors calculations based on data scraped from Upwork.com.

Source: Anwar & Graham (2022)

4.2. Socio-economic similarities (and disparities)

Who are these workers?

Scholars study: education, demographics, entry paths, profiles, earnings, conditions, power relations, etc.

Profile: across platforms, AI workers are

- younger and highly educated relative to national labor markets
- Economically constrained



Source: Tan and Cabato (2023)
Scale AI's Remotasks workers in the Philippines cry foul over low pay »
Washington Post

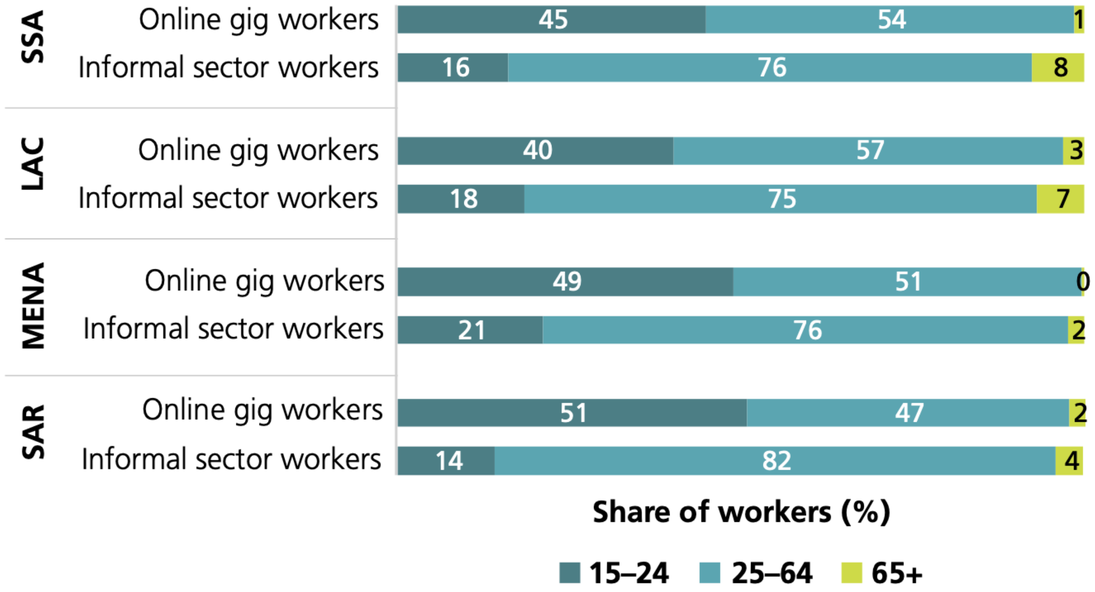
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FIGURE 0.14: Age composition of online gig workers compared to informal workers in labor force surveys, by region



Source: Analysis based on the global survey conducted by the study team and labor force and household surveys (https://ilostat.ilo.org/data/).
Note: LAC = Latin America and the Caribbean; MENA = Middle East and North Africa; SAR = South Asia region; SSA = Sub-Saharan Africa.

Source: World bank (2023)

What do they earn?

- **Piece rates ≠ hourly income** once unpaid time (search, returns, rejections) is counted.
- Typical marketplace worker lands around $1.8–$3.5 per hour (median ≈ $1.77/h, mean ≈ $3.13/h in the MTurk benchmark).
- **Pay varies hugely by tas**k type (e.g., video eval >> transcription/image typing).

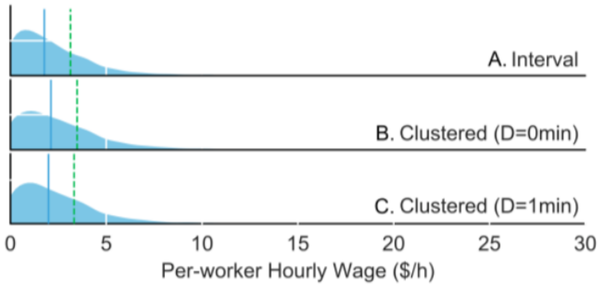


Figure 6. Distributions of per-worker hourly wages based on the interval-based and cluster-based methods. The blue and green lines indicate median and mean.

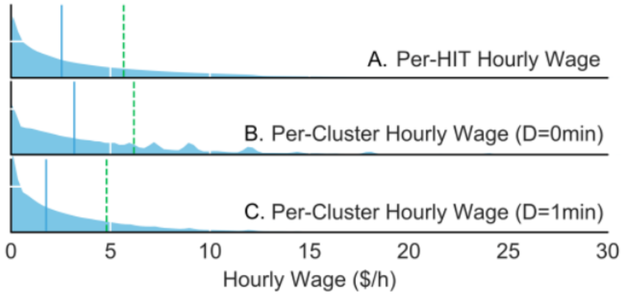


Figure 5. Distributions of per-HIT and per-cluster hourly wages. The blue and green lines indicate median and mean.

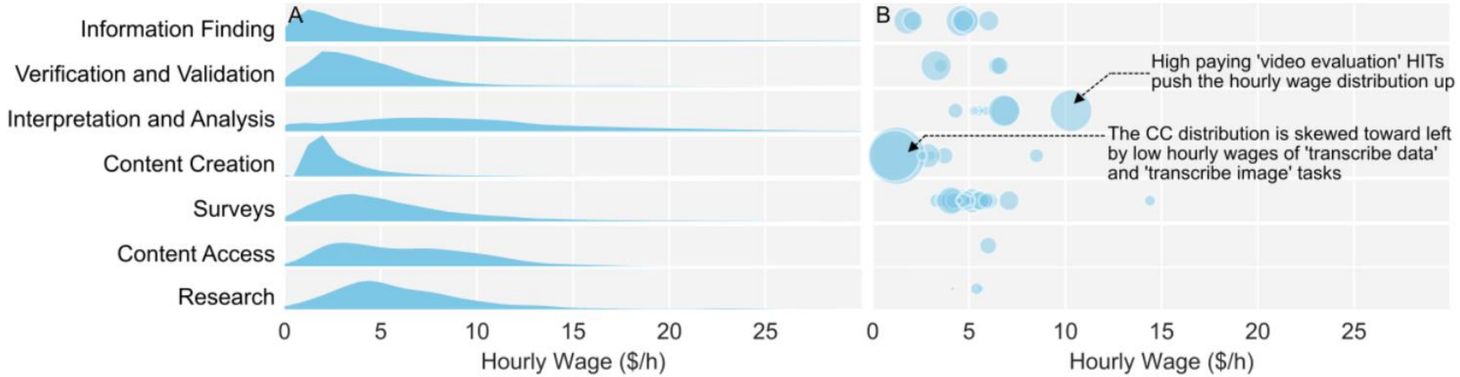
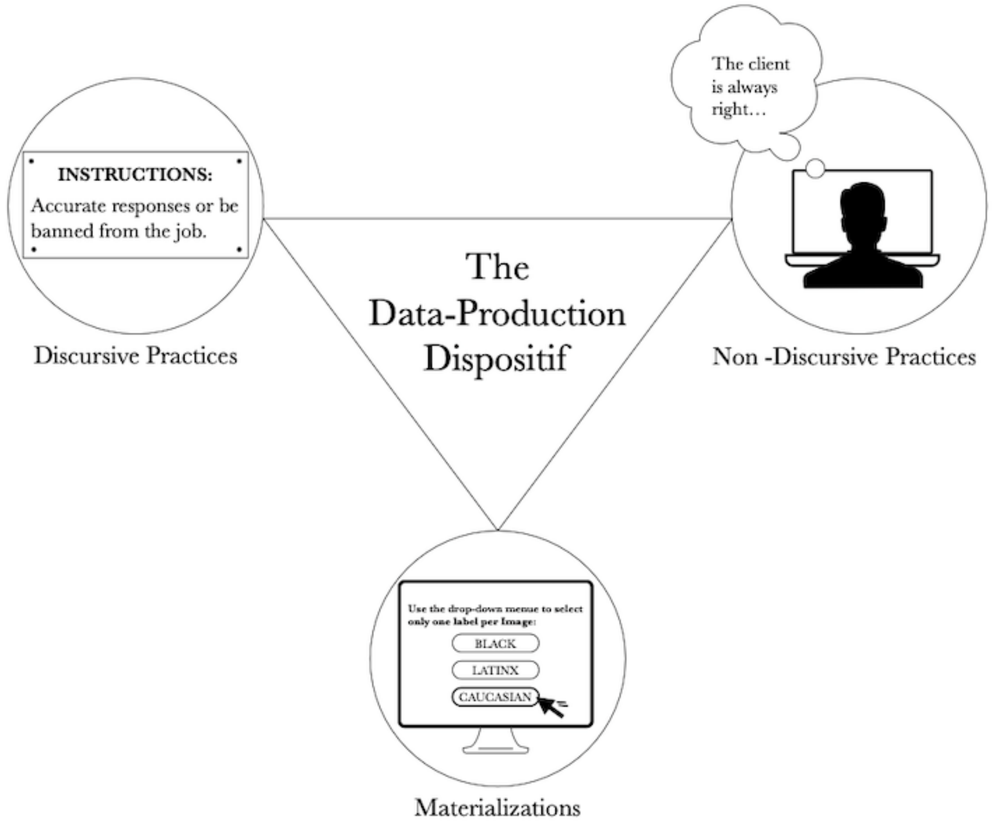


Figure 12. (a) Hourly wage distributions of seven HIT categories provided by Gadiraju *et al.* [25] (with an additional category *Research*). (b) Strip plots showing median hourly wages of HITs associated with the topical keywords in Table 4.

Source: Hara et al. (2018)

Working conditions

- Two settings: **home/atomised marketplaces** (isolation, constant availability, opaque disputes) vs **on-site BPO** (queues/training but tight surveillance & short contracts).
- Instruction–interface–metrics dispositif governs truth and pace (accuracy bars, timers, bans).
- Routine content exposure (violent/sexual) + payment frictions add hidden costs.



The three components of the data-production dispositif
Miceli & Posada (2022)

Gender issue

- **Participation**: women ≈ one-third to ~40%, but composition varies by institutional form (more mixed on marketplaces; often male-skewed entry in some BPOs).
- **Earnings gap** (e.g., ~68% in some LATAM platform surveys) driven by task segmentation, fragmented hours, tooling/network access.
- Where women access scripts, requester groups, and training, the gap narrows.

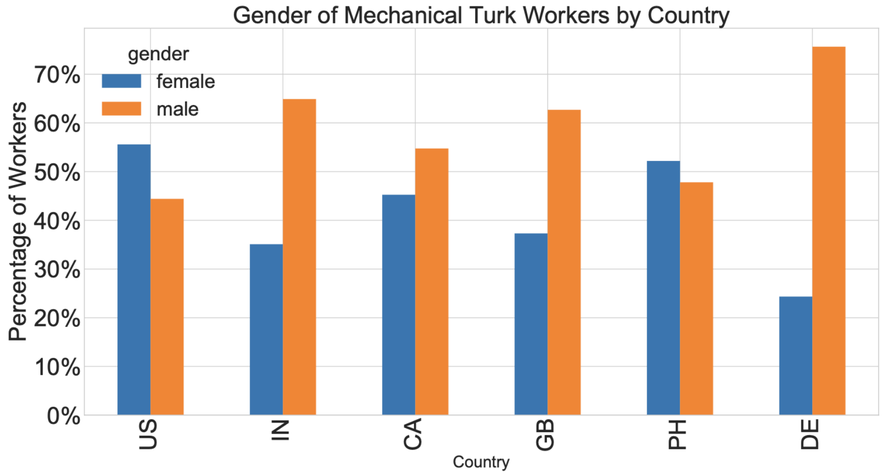
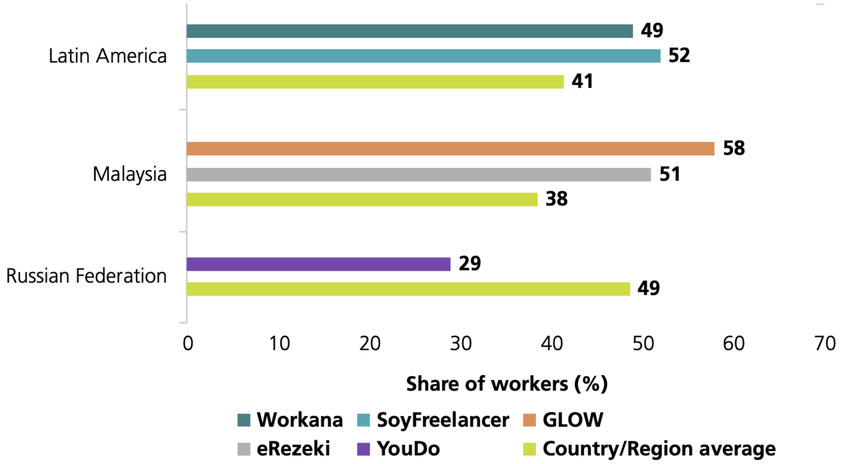


Figure 3: Gender breakdown across countries.

Source: Difallah et al. (2018)

FIGURE 0.15: Women's participation in the labor force and in select online gig work platforms



Source: Analysis based on the platform surveys conducted by the study team.

Source: World bank (2023)

4.3. Case studies: examples of workers in Venezuela and Madagascar

Case study 1: Venezuela (marketplace survivalism)

- **Platform USD as lifeline** amid hyperinflation/sanctions → entry via **marketplaces**; earnings typically **very low** (median hourly ≈ **$0.7–$2** range in your deck's sources), **~15 h/week** typical
- **Infrastructure & payments**: blackouts/slow internet + intermediated cash-out (**fees/delays**) → push workers toward **short, low-bandwidth, lowest-paid** tasks (captchas, selfies, simple clicks)
- **Work organisation**: constant availability, reliance on **Discord/Telegram** groups, **qualification gating**; occasional **quotas** limiting access (e.g., 10% caps)
- **Task mix**: segmentation & classification, **content moderation**, **search relevance**, **data generation** (selfies/voice)

« Most of the time, she cooks… If she asks, I help her, but she knows that most of the time, I'm [working] online. »
Olivia (Tasksource & Workerhub))

« After my mother went to Spain… I couldn't be in a place without Internet… The only solution was to move to [my girlfriend's] house, which is where I live right now. »
Jerónimo (Clickrating))

Posada, J. (2022). *The Coloniality of Data Work: Power and Inequality in Outsourced Data Production for Machine Learning.* PhD dissertation, Faculty of Information, University of Toronto

Case study2 : Madagascar (Francophone BPO chains)

- **Francophone hub** tied to France: AI data work done in **firm-like BPOs** (not open marketplaces), hiring **French-speaking graduates**
- **Contracts & pay**: mix of **journaliers** (day-rated, no paid leave) and short contracts; typical monthly pay bands in MGA shown in your deck (roughly **$0.4–$1.7/hour** equivalents)
- **Work organisation**: **queued assignments**, team leads, **multi-layer QA**, regular syncs with French clients; tighter surveillance but less unpaid search time
- · **Task mix**: classic annotation **+ impersonation** (e.g., **live retail theft detection** from video streams) and continuous dataset updates (canteen/checkout changes)

« We're the ones who detect thefts... he said it was all automatic... But that's not true, we're the ones behind it... it's a bit insulting, but... it puts food on the table. » (Nity, data annotator)

« These [data worker] are considered employees, but they do not get vacation time because their wages are based on the number of days worked... The contract becomes permanent once the worker has completed twenty consecutive days of work and six months in the same position. » (Manager)

Le Ludec, C. (2024). *Des humains derrière l'intelligence artificielle. La sous-traitance du travail de la donnée entre la France et Madagascar.* Doctoral thesis, Institut Polytechnique de Paris (Télécom Paris)